Neural Piano Music Generation from a Procedurally Generated Symbolic Dataset

Sai Abhishek Bhattiprolu
APM 527: Deep Neural Networks

Shivam Sharma
APM 527: Deep Neural Networks

Vansh Bharat Gala
APM 527: Deep Neural Networks

***Abstract*—We train a two-layer LSTM language model to generate monophonic piano melodies and compare it with a linear softmax model, a bigram Markov chain and a unigram reference. All training data is synthetic: a seeded rule-based generator produces 6,000 melodies (531,926 event tokens) with controlled keys, rhythms, motifs and cadences. On held-out pieces the LSTM reaches a test cross-entropy of 1.010 nats per token (perplexity 2.75), against 1.687 for the linear model and 2.348 for the bigram chain. Sampled at temperature 1.0, 98.0% of LSTM sequences are grammatically valid and 91.7% fill a whole number of bars, compared with 16.0% for the linear model. The LSTM reproduces local pitch, interval and rhythm statistics closely but ends on the tonic less often than the training data and repeats motifs far less, so phrase-level form is only partly learned. A selected 32-second sample is exported as a MIDI file.**

I. Introduction

Recurrent networks have been used for symbolic melody generation since early LSTM work on blues improvisation [2]; event-based token representations later made the task a standard next-token prediction problem [3], [4]. We ask a narrow question: *can a compact LSTM trained only on procedurally generated melodies learn their local melodic and rhythmic structure well enough to (a) predict held-out pieces better than linear and Markov baselines and (b) sample new, valid, bar-aligned melodies?* Because the data comes from known rules, we can measure how far generated sequences depart from them. The experiment says nothing direct about human-composed music, and no listening study was run.

II. Procedural Dataset

Melodies are in 4/4 with a time unit of one 16th note. Each piece draws a tonic (12 choices) and a mode (major or natural minor, with a raised leading tone at the final cadence). Pitches come from a scale-degree random walk: steps of ± 1 degree have probability 0.27 each, ± 2 0.12 each, repeated notes 0.06, and leaps of ± 3 or ± 4 degrees 0.14 in total; after a leap the next step reverses direction with probability 0.7. Notes are kept within MIDI 48–84. Each bar uses one of 13 rhythm templates (eighth to half notes, dotted values, quarter and half rests), so bars are always filled exactly. A one-bar motif (25% of them arpeggiated) opens each piece and is repeated a step or two higher; contrasting phrases use arpeggios on I, IV, V or vi. Pieces have 2, 3 or 4 four-bar phrases; inner phrases end on the dominant and the final phrase restates the motif and ends on the tonic.

Each piece is encoded as `START`, then `NOTE_`p `DUR_`d for a note of pitch p and length $d \in \{2, 4, 6, 8, 12, 16\}$, or `REST_`$d$ with $d \in \{4, 8\}$, then `END`. With `PAD`, the vocabulary has $V{=}48$ tokens. The seeded generator produced 6,000 pieces (2,089, 1,969 and 1,942 of 8, 12 and 16 bars; mean length 88.7 tokens). A transposition-invariant signature (mode, interval sequence, rhythm) found no duplicates. We split by piece into 4,800 / 600 / 600 for training, validation and test (425,405 / 52,813 / 53,708 tokens), and checked that no signature appears in two splits. The data is synthetic and inherits the generator's biases: it is diatonic, monophonic, metrically rigid and has no dynamics.

III. Models and Training

All models predict the next token given the preceding ones and are evaluated on the same test targets. *Unigram* and *bigram* models use add-α counts, with α chosen on validation (1.0 and 10^{-3}). The *linear* baseline, following the course guideline to compare against a linear model, is a multinomial logistic regression on the one-hot encodings of the previous $K{=}8$ tokens,

$$p(x_{t+1} \mid x_{\le t}) = \mathrm{softmax}\Big(b + \textstyle\sum_{k=0}^{K-1} W_k\, e_{x_{t-k}}\Big), \quad (1)$$

with $W_k \in R^{V \times V}$ (18,480 parameters). The *LSTM* [1] embeds tokens in 64 dimensions, applies two LSTM layers of 192 units with dropout 0.2 [5] between and after them, and projects to V logits (506,928 parameters). Both neural models minimise token-level cross-entropy (padding ignored) with Adam [6], batch size 64, gradient clipping at 1.0, and learning rates $2{\times}10^{-3}$ (LSTM) and $5{\times}10^{-3}$ (linear). Training runs for at most 40 epochs with early stopping (patience 3) on validation loss, and the checkpoint with the lowest validation loss is kept. Test loss is recorded every epoch only to plot it; it plays no part in model selection. On a 2-core CPU an LSTM epoch took about 13.5 s.

IV. Evaluation Protocol

We report test cross-entropy $\mathcal{L}$ in nats per predicted token and perplexity $e^{\mathcal{L}}$. For generation we draw 300 sequences per model by ancestral sampling with temperature T (no top-k truncation; `PAD` and `START` masked; at most 400 tokens) and apply no grammar constraint. Metrics: *sequence validity* (no token violates `START (NOTE DUR | REST)* END`); *bar fit* (total length is a whole number of bars); *scale adherence* (share of notes in the best-fitting major or minor scale); *tonal closure* (final note is the tonic of a scale containing $\geq 90\%$ of the notes); total-variation (TV) distance between generated and

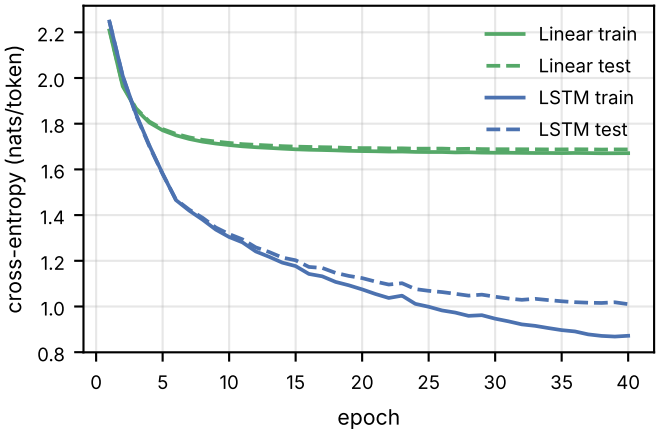


Fig. 1. Per-epoch cross-entropy on the training and test sets (evaluation mode).

TABLE I
NEXT-TOKEN PREDICTION ON THE TEST SET (53,108 TARGETS)

Model	Params	CE (nats)	Perplexity
Unigram (reference)	46	3.079	21.73
Bigram Markov	2,208	2.348	10.46
Linear, K=8	18,480	1.687	5.40
LSTM	506,928	**1.010**	**2.75**

test interval histograms; and *repeat-4*, the share of repeated (interval, duration) 4-grams within a piece, which reflects motif reuse. These are proxies for rule conformity, not for musical quality.

V. RESULTS

Fig. 1 shows the training and test curves required by the course. The linear model levels off after about 10 epochs at a test loss near 1.69, with almost no train–test gap. The LSTM improves more slowly; over the last four epochs its validation loss stayed between 1.024 and 1.034 while training loss kept falling, and the final train/validation/test losses are 0.872/1.024/1.010. Training reached the epoch cap before early stopping triggered, so epoch 40 was selected. Table I gives test results: the LSTM lowers perplexity from 5.40 (linear) to 2.75.

Table II separates what each model captures. The bigram chain almost always obeys the note–duration grammar, a first-order constraint, but its absolute-pitch transitions produce wrong interval statistics (TV 0.478) and it rarely fills bars. The linear model matches interval statistics but cannot track position within the bar (bar fit 0.160). The LSTM keeps 91.7% of samples bar-aligned at T=1.0, and on average 98.0% of a sample's notes lie in one scale. Lower temperature raises validity and tonal closure; higher temperature lowers both. Two gaps remain at every temperature: tonal closure stays well below the data (0.54 vs. 1.00 at T=1.0), and motif repetition is much lower than in the data (0.002 vs. 0.045). No sample from any model was an exact copy of a training piece.

TABLE II
STATISTICS OF 300 SAMPLED SEQUENCES PER MODEL (FRACTIONS)

Source	Valid	Bar fit	Scale	Closure	TV int.	Rep-4
Test data	1.000	1.000	1.000	1.000	0.000	0.045
Bigram, T=1	0.990	0.130	0.812	0.031	0.478	0.038
Linear, T=1	0.877	0.160	0.869	0.153	0.025	0.033
LSTM, T=0.7	0.993	0.990	0.992	0.667	0.062	0.006
LSTM, T=1.0	0.980	0.917	0.980	0.540	0.029	0.002
LSTM, T=1.3	0.853	0.742	0.967	0.445	0.044	0.001

For the music artifact we fixed a selection rule before listening: sample LSTM candidates at T=1.0 with seed 2026 and keep the first that is grammatically valid, bar-aligned, 30–60 s long at 90 BPM, and not a training copy. The first candidate passed (23 of 50 did). It was written to `generated_piano.mid` (Acoustic Grand Piano, 90 BPM, velocity 80, notes held for 95% of their value): 51 notes, 31.9 s, MIDI 62–84, with no overlapping or zero-length notes, plus a FluidSynth WAV rendering. Both are qualitative examples only.

VI. DISCUSSION AND LIMITATIONS

The LSTM's advantage over the linear model is largest on metrical structure, which needs a running count of elapsed time that a fixed window of eight tokens cannot hold. Its weaknesses are long-range: returning to the opening motif and closing on the tonic depend on context from many bars earlier. Since the train–validation gap widens after epoch 20, bar-position tokens or attention seem more promising than extra capacity. Limitations: the data is synthetic and simple, so results do not transfer to real piano repertoire; there is no polyphony, dynamics or expressive timing; each model was trained once with one seed, so run-to-run variance is unmeasured; and all metrics measure conformity to the generator, not musical value.

VII. CONCLUSION

On procedurally generated melodies, a 0.5 M-parameter LSTM clearly outperforms linear and Markov baselines in held-out likelihood and produces mostly valid, bar-aligned, diatonic samples. It captures local rules well but reproduces phrase-level form only partly. All data, models, metrics and the generated MIDI are produced by one seeded Python script.

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